Smart Lab Portal

Project report: core system, features, results and status

Thesis: *Design and Development of an Intelligent IoT-Based Smart Research Laboratory Management System*. BSc Robotics & Automation Engineering, German International University (GIU), Cairo.

Author: Ali Loay **Snapshot:** 29 October 2026, 17:10

Labels used for every result: **verified** (checked in code or tests), **live-verified** (checked on the running portal), **simulated** (browser device emulation, not a real phone), **unverified** (not yet measured).

1. The project in one page

Smart Lab is one local system that runs a university laboratory building: booking, two-factor door access, the robots, PCs, the physical environment, consumables, reusable equipment, security cameras, automation and AI assistants. Everything runs on the local **Smart_Lab** network - no cloud dependency, no internet needed.

The portal manages **11 laboratories** (LAB_01 - LAB_11). The pilot lab with all hardware is **LAB_01 Robotics Lab**.

Code	Laboratory	Code	Laboratory
LAB_01	Robotics Lab (pilot)	LAB_07	Microcontrollers Laboratory
LAB_02	PLC Lab	LAB_08	Power Lab
LAB_03	Production Lab	LAB_09	Design and Fabrication Lab
LAB_04	Materials Lab	LAB_10	CAD & Digital Design Laboratory
LAB_05	Automotive Lab	LAB_11	Fashion Design & Textiles Lab
LAB_06	PCB Lab		

Thesis numbering: System 1 = core robotics, PCs and LAN; System 2 = physical environment; System 3 = consumables; System 4 = reusables. The door is a separate, protected subsystem.

2. Architecture

Layer	Technology	Runs on
Portal (web app)	React 18, TypeScript, Tailwind; nginx	Docker: frontend
API	FastAPI (Python), 31 route modules	Docker: backend
Database	PostgreSQL 16, nightly backups	Docker: db, backup
Automation	n8n, 12 workflows (never touches the door)	Docker: n8n
Door	Master ESP32 + ESP32-CAM + face server (SFace)	Firmware; laptop port 5000
Security cameras	YOLO26s person detection (DirectML, GTX 1650), tracking, zones	Laptop ports 5100, 5101
AI assistants	Ollama qwen2.5:1.5b on the GPU + knowledge agent	Laptop port 5199
Voice	faster-whisper speech-to-text + Piper text-to-speech	Laptop port 5320
Robots	UR5e collector (ur_rtde, read-only), FANUC 3D twin	tools/robot_monitor
IoT nodes	ESP32 environment node, consumables bin, reusable bin	firmware/
Bin detector (new)	YOLO26n, ONNX Runtime on the CPU (GPU stays with cameras)	Inside the backend
Launcher	launch.bat starts the whole stack in one click	Repository root

Size of the codebase: 384 commits, 39 portal pages, about 670 backend test functions (719 test cases with parameters), 42 documentation files, 7 firmware projects.

3. Subsystems and features

3.1 Door - protected two-factor access

- Step 1: booking QR code (valid only for one lab and inside the booking window) or RFID card.
- Step 2: fingerprint or face recognition (SFace templates on the laptop face server).
- Session trace: every booking reconstructs to the second - credential, both factors and identities, result, entry, door cycle; an exit only when one is actually observed.
- Enrolment from the portal: fingerprint, face (also from uploaded photos), RFID cards; each account gets its door identity (USER3, USER4, ...) automatically.
- Safety rule: only the master ESP32 opens the door; no software path, automation or AI can unlock it.

3.2 System 1 - robots, PCs and LAN

- UR5e: live collector (six joints, velocity, TCP, robot/safety modes; LIVE / STALE / OFFLINE) and 3D digital twin.
- Robot control routes exist, but every adapter stays UNAVAILABLE until the owner signs off a commissioning step. Nothing in the system moves a robot.
- FANUC: 3D twin and equipment record.
- Lab PCs: pc_agent reports each PC's state; the lab engineer controls the lab PCs.

3.3 System 2 - physical environment

- ESP32 environment node: temperature, humidity, flame and further sensors, live in the portal.
- Anomalies from rules, sensor-fault checks and robust statistical baselines (median / MAD) - explainable, no black-box model.

3.4 System 3 - consumables

- Load-cell bin for resistors, capacitors and LEDs; stock tracked by weight and by what the student declares.

3.5 System 4 - reusables

- Smart bin for Arduinos, ESP boards and breadboards: NFC tag on each board, NFC reader and LCD outside, motorised lid, load cell inside. Tracked three ways: portal declaration, weight and NFC.
- 3D-printed enclosures designed for all four boxes (door unit, environment node, consumables bin, reusable dock).

3.6 Security cameras

- Two cameras in LAB_01, one process each (a failing camera never stops the other).
- Person detection and tracking; names from the door's face match, only on a clear match.
- Equipment zones: a moved or tilted PC is flagged; cross-camera naming.

3.7 Monitoring, operations and maintenance

- Command Center and wall monitor: every lab's door, lock, who is inside by role, alarms.
- Operations Center: utilisation heatmaps, access funnel, refusal reasons, device outages, maintenance against SLA, environment and automation status - all counted from recorded rows, explicit empty states.
- Maintenance workflow: report a problem with photos in about a minute; staff triage, assign, fix and record it.
- Equipment lifecycle: serial number, holder, inspections, next maintenance (overdue flagged), full timeline.

3.8 Phone camera (Scan page)

- QR markers on every asset open a live card (robot joints, bin stock, environment, door, PC).

- People boxes from the camera service and names only on a clear face match.
- "Which box is this?": recognises the enclosures by their look; the new detector covers the reusable bin from far away, from the side and with the QR code turned away.
- Voice questions with live captions, glass controls on the picture, front / back camera, refresh, pause.

3.9 AI assistants and automation

- Staff assistant: reports, usage, recurring faults with fixes, charts, CSV / PDF export. Student helper: bookings, free labs, door refusals. Both answer from the portal's own records through read-only tools; local and free.
- Knowledge agent: answers questions about the project itself (vault, search, graph).
- n8n: 12 workflows - reminders, door-denial bursts, offline escalation, reports, data quality, anomalies.

4. Work in this session (28 - 29 October 2026)

4.1 Every device and every page

Click-through of 48 pages x 3 roles (admin, staff, student) with every safe button, tab and select pressed. **Simulated** in Chrome device emulation.

Device	Pages	Result
iPhone SE (small phone)	48	pass
iPhone 15 Pro Max	48	pass
Pixel 7 (Android)	48	pass
Galaxy Tab S4 (Samsung tablet)	48	pass - two pauses of about 0.2 s
iPad Pro 11 landscape	48	pass
Laptop 1366 x 768	48	pass - two pauses of about 0.2 s
Monitor 2560 x 1440	48	pass

No crashes, blank pages or errors; pages opened in about 0.1 - 0.6 s. Responsive audit (8 profiles): sideways overflow 12 -> 0, clipped elements 29 -> 0, letter-by-letter stacking fixed.

4.2 Tools and memory

- claude-mem repaired (worker path, search add-on, embedding model) and verified: saves and finds memories.
- MCP audit: added Chrome DevTools 1.12.0 (headless, isolated profile, no usage statistics) and Context7 4.4.4, both tested on the portal; 47 unused connectors switched off for this project (reversible).
- n8n: the n8n Cloud workspace no longer exists (404 "No instance here"); the local n8n 2.40.7 has MCP switched off - needs the owner's token.

4.3 Reusable-bin detector

Data: 113 owner photos of the bin (home: tables, three carpets, tiles; near and far), 27 earlier close-ups, and no-bin pictures: background patches, 450 COCO photos and 237 lab-camera frames from before the bin existed. Photos stay on the laptop, metadata (GPS) removed, nothing committed. Boxes drawn automatically (YOLO-World, then the first trained model) and every box checked by eye on contact sheets.

Test: 15 bins in two whole scenes never trained on (coffee table, striped carpet) and 429 no-bin pictures (8 patches and old fooling pictures, 200 COCO photos, 221 lab frames).

Model	Found @0.5	False @0.5	Found @0.6	False @0.6	Found @0.7	False @0.7
Round 2 YOLO11n	14/15	24/429	13/15	3/429	11/15	0/429
Round 2 YOLO26n	9/15	0/429	5/15	0/429	2/15	0/429
Round 3 YOLO11n	11/15	4/429	9/15	1/429	7/15	0/429
Round 3 YOLO26n (interim, 14:15-17:00)	14/15	13/429	14/15	5/429	9/15	0/429
Round 4 YOLO11n (DEPLOYED, conf 0.4)	**14/15**	**0/429**	**14/15**	**0/429**	11/15	0/429
Round 4 YOLO26n	3/15	0/429	1/15	0/429	0/15	0/429

Winner: **round-4 YOLO11n** (round 3 + 237 lab-camera frames as no-bin examples). At 0.3 - 0.6 it finds 14/15 with 0/429 false bins; the highest score on any no-bin picture is 0.19 (round-3 YOLO26n: 0.62). Deployed at **0.4**. **Live-verified** in the running backend: 14/15 bins, 0/44 false, 86 ms per check, bin confidences 0.63 - 0.88. Its outputs match the training library (confidence within 0.022, same call 23/23).

The detector only runs when the photo matcher finds nothing, and its answer is a hint: identity and live data still come from the QR marker.

4.4 Speed

What	Before	After	How
Bin detector per frame	525 ms (OpenCV DNN)	86 ms warm (live)	ONNX Runtime CPU, 2 threads
Photo matcher per frame (p50)	421 ms	333 ms	28 reference photos on 4 cores, same decisions
First "which box?" after restart	+ about 6 s	no wait	warm-up at backend start
Phone-local YOLO26s (rejected)	-	2.7 s / 13.4 s per frame	simulated browser; server: about 37 ms

4.5 Tests

- Backend: 719 passed, 2 skipped, 0 failed (full suite, test database only).
- New tests: detector letterbox / box mapping, threshold, missing model, OpenCV fallback; route PROBABLE via detector and UNKNOWN without it.
- Frontend: type check clean, production build passes.
- Commits (local only, not pushed): 7fe4e9d, a962057, 7075d8e, 8059f78 and earlier 62117c7, 7d66d27, 6239ce1.

4.6 Incidents

- Windows Update restarted the laptop at 08:37 and 13:54 on 29 October (event 1074). Round-4 training stopped at epoch 28 and the portal was offline until about 14:15.
- Recovered: Docker and all lab services restarted with launch.bat; training resumed from epoch 28 as a separate process; Windows Update paused until 5 November 2026.

5. Current status

Area	Status
Portal, booking, accounts, monitoring	Live
Door (two-factor)	Validated at the lab
Environment node	Live in the portal
Consumables and reusable bins	Built; enclosures designed and printed
Security cameras	Services running; streams offline while away from the lab
UR5e	Read-only collector ready; commissioning needs the owner at the lab
Bin detector	Live (round-4 YOLO11n, conf 0.4): 14/15 unseen, 0/429 false
AI assistants and voice	Running
Training rounds 2 - 4	Done: six models compared, best deployed 17:00
Devices	7 profiles pass (simulated); real iPhone / Android test outstanding

6. Open items

- Done: round 4 finished, six models compared, best deployed and live-verified.
- Owner phone test of the bin detector (far, side, QR turned away).
- Outdoor photos of the bin (street, car, roof, pool; sun and shade): it has only seen indoor scenes, so "anywhere" is not proven yet.
- Photo sessions for the full lab model: reusable bin (done), consumables bin, master ESP box, environment box, UR5e, FANUC, lab PCs, door, lab engineer desk, TA desk, security cameras 1 and 2, lab in general, robot controllers, network devices, PLC (optional). 60 - 100 photos or a 1 - 2 minute video each; Colab notebook ready.
- n8n: switch on MCP access in the local n8n and send the token.
- Change the seed accounts' passwords; UR5e commissioning; long run with both cameras.

7. Hardware proposals (end of project, not now)

Option	Rough price	Use
Depth camera (Intel RealSense / Luxonis OAK-D)	-	Measured distance for robot safety zones
Orange Pi 5 (RK3588, 6 TOPS)	about $70 - 100	Real-time YOLO for extra cameras, off the laptop GPU
Luckfox Pico (RV1103/1106)	about $15 - 30	Cheapest per-camera YOLO-nano, low resolution
Raspberry Pi 5 + AI HAT	about $130 - 180	Easiest, most expensive
Raspberry Pi Pico	-	Not suitable for vision (264 KB RAM); ESP32 already covers I/O

Prices are rough and unverified. Software estimates of depth are never used as safety data; the system is not a certified industrial safety system.